

Degree Growth of **H**olonomic **P**olynomial **S**equences and Applications

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Polynomial Sequences

Definition. Let R be an integral domain.

$$\begin{aligned} P: \mathbb{N} &\longrightarrow R[x] \\ n &\longmapsto P_n(x) \end{aligned}$$

Examples.

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Examples.

1. Orthogonal polynomials.

- ▶ Laguerre polynomials: $L_0(x) = 1$, $L_1(x) = 1 - x$,

$$L_n(x) = \sum_{k=0}^n \frac{(-1)^k}{k!} \binom{n}{k} x^k.$$

$$(n+1) \cdot L_{n+1}(x) = (2n+1-x) \cdot L_n(x) - n \cdot L_{n-1}(x).$$

- ▶ Chebyshev, Jacobi, Hermite, Gegenbauer, Legendre, ...

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Examples.

2. Polynomials in combinatorics.

- ▶ Euler polynomials: $E_0(x) = 1$, $E_1(x) = x - 1/2$,

$$\frac{2\exp(xt)}{\exp(t) + 1} = \sum_{n=0}^{\infty} E_n(x) \frac{t^n}{n!} \quad \Rightarrow \quad E'_n(x) = nE_{n-1}(x)$$

- ▶ Bernoulli, Bell, Genocchi, Stirling, ...

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3. Polynomials in geometry.

Definition. X : topological space with f. g. homology group. The **Poincaré polynomial** $P_X(x)$ is the generating fun. of **Betti** numbers.

Example. Poincaré polynomial of compact Lie groups

$$P_{SU(n+1)}(x) = (1+x^3)(1+x^5)\cdots(1+x^{2n+1})$$

$$P_{SO(2n+1)}(x) = (1+x^3)(1+x^7)\cdots(1+x^{4n-1})$$

...

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4. Polynomials in algebra.

Definition. $V = \bigoplus_{i \in \mathbb{N}} V_i$: $\mathbb{N}$ -graded vector space over K with $\dim_K(V_i) < +\infty$. The n th **Poincaré polynomial** of V is

$$P_{n,V}(x) = \sum_{i=0}^n \dim_K(V_i) x^i.$$

Example. Let $V = K[x_1, \dots, x_m]$ and $V_i = \{p \in V \mid \deg(p) \leq i\}$. Then

$$P_{n,V}(x) = \sum_{i=0}^n \binom{m+i}{i} x^i \quad \Rightarrow \quad P_V(x) = \frac{1}{(1-x)^{m+1}}.$$

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Examples.

4. Polynomials in number theory.

Definition. Let $\alpha_n \in \overline{\mathbb{Q}}$ and $P_n(x)$ be the minimal polynomial of α_n .

Example. Let $\alpha = \sqrt{2}$ and $\alpha_n = \alpha^n$. Then

$$\alpha_1 = \sqrt{2} \qquad P_1 = x^2 - 2$$

$$\alpha_2 = 2 \qquad P_2 = x - 2$$

$$\alpha_3 = 2\sqrt{2} \qquad P_3 = x^2 - 8$$

$$\vdots$$
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Problems on Polynomial Sequences

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Problem 1. Degree growth. Determine $d_n := \deg_x(P_n(x))$.

- ▶ Orthogonal polynomials: $d_n = n$

$$P_n(x) = (A_n x + b_n)P_{n-1}(x) + C_n P_{n-1}(x).$$

- ▶ Bernoulli polynomials: $d_n = n$

$$B_0 = 1, \quad B_1 = x - \frac{1}{2}, \quad B'_n(x) = nB_{n-1}(x).$$

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Problem 1. Degree growth. Determine $d_n := \deg_x(P_n(x))$.

► Minimal polynomials for α^n : if $\alpha = \sqrt{2}$, then

$$d_n = \begin{cases} 2, & n \text{ is odd;} \\ 1, & n \text{ is even.} \end{cases}$$

Conjecture. For any $\alpha \in \overline{\mathbb{Q}}$, $d_n := \deg_x(P_n(x))$ is periodic, where $P_n(x)$ is the minimal polynomial of α^n .

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Problem 2. Generating functions. Determine

$$f(x, y) := \sum_{n=0}^{\infty} P_n(x) y^n.$$

$$\sum_{n=0}^{\infty} L_n(x) y^n = \frac{\exp(-xy/(1-y))}{1-y}.$$

Problem 3. Zero distribution. Determine the zeros of $P_n(x)$.

Are all of the zeros of $P_n(x)$ real for any n ?

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Problem 4. Coefficient distribution. Are the coefficients of $P_n(x)$ unimodal, log-convex, log-concave?

Problem 5. Irreducibility. For which n , $P_n(x)$ is irreducible?

Question. Is Euler polynomial $E_n(x)$ irreducible if n is odd and $E_n(x) = x \cdot \bar{E}_n(x)$ with $\bar{E}_n(x)$ irreducible if n is even?

$\vdots$

Holonomic Polynomial Sequences (HOPS)

Definition. A polynomial sequence $P_n(x)$ is **holonomic** if

$$A_d \cdot P_{n+d}(x) + A_{d-1} \cdot P_{n+d-1}(x) + \cdots + A_0 \cdot P_n(x) = 0,$$

where $A_i \in R[n, x]$ and $A_d \cdot A_0 \neq 0$. If $A_i \in R[x]$, $P_n(x)$ is **C-finite**.

Examples.

- ▶ Orthogonal polynomials:

$$T_{n+1} - 2xT_n(x) + T_{n-1}(x) = 0$$

$$P_n(x) = (A_n x + B_n)P_{n-1}(x) + C_n P_{n-1}(x)$$

- ▶ Poincaré polynomial sequences are **often** C-finite.
- ▶ Euler, Bernoulli etc. polynomial sequences are **not** holonomic.



Beer and Hops

Degree Growth of HOPS

Problem. Given a HOPS $P_n(x)$, determine $d_n := \deg_x(P_n(x))$.

Definition. d_n is **uniformly bounded** if $\exists M \in \mathbb{N}$ s.t.

$$d_n \leq M \text{ for all } n \in \mathbb{N}.$$

Remark. d_n grows at most linearly and is uni. bounded if $A_i \in R[n]$.

Definition. A nonempty set $S \subseteq \mathbb{N}$ is **syndetic** if $\exists C \in \mathbb{Z}_+$ s.t.

$$n \in S \Rightarrow n+i \in S \text{ for some } i \in \{1, \dots, C\}.$$

Example. $\{2n \mid n \in \mathbb{N}\}$ is syndetic, but $\{p_1^{m_1} \cdots p_n^{m_n} \mid m_i \in \mathbb{N}\}$ with p_i prime numbers is not syndetic.

Degree Growth of HOPS

Theorem. Let $P_n(x)$ be a HOPS and $d_n := \deg_x(P_n(x))$. Then either d_n is uniformly bounded or $\exists \kappa \geq 1/d$, $B \in \mathbb{N}$, and a syndetic $S \subseteq \mathbb{N}$ s.t.

$$d_n \geq \kappa n - B \quad \text{for all } n \in S.$$

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Remark. κ can reach $1/d$, e.g., $P_n(x) = xP_{n-d}(x)$ for $n \geq d$ and $P_0 = \cdots = P_{d-2} = 0$ and $P_{d-1} = 1$. Then $d_n = n/d - (d-1)/d$ if $n \equiv -1 \pmod{d}$, otherwise $d_n = 0$.

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Conjecture. Let $P_n(x)$ be a hops and $d_n := \deg_x(P_n(x))$. Then $\{d_n\}_{n \geq 0}$ is a **Skolem** set, i.e., a union of finitely many arithmetic progressions.

Remark. This conjecture is **hard** since it implies **Rubel's conjecture** (1983) on P-recursive sequences.

Exponential Sums over Finite Fields

Let p be a prime and ζ_p be the primitive p -th root of unity.

Definition. For $f \in \mathbb{F}_p[x_1, \dots, x_n]$, the **exponential sum** of f is

$$S_k(f) := \sum_{(x_1, \dots, x_n) \in (\mathbb{F}_{p^k})^n} \zeta_p^{\text{Tr}_k(f(x_1, \dots, x_n))} \in \mathbb{Z}[\zeta_p],$$

where $\text{Tr}_k : \mathbb{F}_{p^k} \rightarrow \mathbb{F}_p$ is the trace map.

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Wan's Conjecture. Let $P_k(x)$ be the minimal polynomial of $S_k(f)$. Then $d_k := \deg_x(P_k(x))$ is **eventually periodic**, i.e.,

$$\exists M, T \in \mathbb{N} \text{ s.t. } d_{n+T} = d_n \text{ for all } n \geq M.$$

Remark. The above d_k is uniformly bounded by $p-1$.

Degree Growth of Algebraic Sequences

Theorem. Let $\alpha_n \in K$ with $[K : \mathbb{Q}] < +\infty$ be **C-finite** and $P_n(x)$ be the minimal polynomial of α_n . Then

- ▶ $d_n := \deg_x(P_n(x))$ is eventually periodic.
- ▶ $P_n(x)$ is C-finite.

Corollary. Let $\alpha \in \overline{\mathbb{Q}}$ and $P_n(x)$ be the minimal polynomial of α^n . Then $P_n(x)$ is C-finite and $\deg_x(P_n(x))$ is eventually periodic.

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Corollary. Wan's conjecture is true.

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Proof. By Dwork-Bombieri-Grothendieck rationality theorem

$$L(f, T) := \exp \left(\sum_{k=0}^{\infty} \frac{S_k(f)}{k} T^k \right) \in \mathbb{C}(T).$$

Separability Problem and Degree Growth

Problem. Decide whether $f(t, \mathbf{x})$ with $\mathbf{x} = \{x_1, \dots, x_n\}$ satisfies

$$L(t, \partial_t)(f) = 0, \quad \text{where } L \in \mathbb{F}(t)\langle \partial_t \rangle \setminus \{0\} \text{ with } \text{char}(\mathbb{F}) = 0.$$

If L exists, say f is ∂_t -separable.

Example. $f = \sqrt{t(x^2 + 1)} + 1$ is D_t -separable since

$$L(f) = 0, \quad \text{where } L = 2t \cdot D_t^2 + D_t.$$

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Applications.

- ▶ Separation of variables in PDE:

$$\frac{\partial y}{\partial t} - c \frac{\partial^2 y}{\partial x^2} = 0 \quad \rightsquigarrow \quad \frac{\partial y}{\partial t} - \lambda y = 0 \quad \text{and} \quad c \frac{\partial^2 y}{\partial x^2} - \lambda y = 0.$$

- ▶ Picard–Fuchs equations for differential forms;
- ▶ Existence problems in creative telescoping;
- ▶ Holonomic polynomial sequences.

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Theorem. Let $P_n(x)$ be a HOPS. Then

$d_n := \deg_x(P_n(x))$ is uniformly bounded



$P_n(x)$ is S_n -separable, i.e., $\exists L \in \mathbb{F}[n]\langle S_n \rangle$ s.t. $L(n, S_n)(P_n(x)) = 0$.

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Degree Growth and Vanishing Sums

Theorem (Nicole1717) Let $P_n(x) \in \mathbb{C}[x]$ with $\deg_x(P_n(x)) \leq n-2$ and

$$f(x) = \frac{P_n(x)}{(x + \beta_1) \cdots (x + \beta_n)} = \sum_{i=1}^n \frac{\alpha_i}{x + \beta_i},$$

where $\beta_i - \beta_j \in \mathbb{Z} \setminus \{0\}$ for $i \neq j$. Then $\sum_{i=1}^n \alpha_i = 0$.

Corollary. Let $P_n(x) \in \mathbb{C}[x]$ with $\deg_x(P_n(x)) \leq n-1$. Then

$$\sum_{k=0}^n \frac{P_n(-k)}{(-1)^k k! (n-k)!} = 0.$$

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Proof.

$$f(x) = \frac{P_n(x)}{x(x+1) \cdots (x+n)}.$$

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Example. Taking $P_n(x) = n!x^j$ with $0 \leq j < n$,

$$f = \frac{n!(-x)^j}{x(x+1) \cdots (x+n)} \rightsquigarrow \sum_{k=0}^n (-1)^k \binom{n}{k} k^j = 0.$$

Gould's Vanishing Sums

Example (*Combinatorial Identities*, p. 63).

$$\sum_{k=0}^{2n+1} a(n,k) = 0, \quad \text{where } a(n,k) = (-1)^k \binom{2n+1}{k} \binom{2k}{k} \frac{1}{\binom{n+k+1}{k}}.$$

Consider

$$\sum_{k=0}^{2n+1} \frac{a(n,k)}{x+k} = \frac{P_n(x)}{x(x+1) \cdots (x+2n+1)}.$$

By **creative telescoping**, we get

$$b_3 P_{n+3}(x) + b_2 P_{n+2}(x) + b_1 P_{n+1}(x) + b_0 P_n(x) = 0,$$

where $b_3, b_2, b_1, b_0 \in \mathbb{Q}[n, x]$ with

$$(\deg_x(b_3), \deg_x(b_2), \deg_x(b_1), \deg_x(b_0)) = (2, 4, 6, 5).$$

$$(\deg_x(P_1), \deg_x(P_2), \deg_x(P_3)) = (2, 4, 6) \Rightarrow \deg_x(P_n(x)) \leq 2n$$

Summary



Holonomic Polynomial Sequences (HOPS)

Thanks!